

Repo structure

Backend Structure

```
pvt-app/
|
|— go.mod                                # Go module root
(github.com/durelius/pvt-app)
|— go.sum
|
|— backend/
|   |
|   |— api-gateway/                      # Reverse proxy – single entry point
|   (:8080)
|   |   |
|   |   |— .air.toml                    # Hot-reload config (air)
|   |   |— app/
|   |   |   |— main.go                 # CORS middleware, dynamic reverse
|   |
|   |— proxy to services
|   |   |
|   |   |— services/
|   |   |   |
|   |   |   |— auth/                   # Auth service placeholder (:8081)
|   |   |   |   |
|   |   |   |   |— .air.toml
|   |   |   |   |— app/
|   |   |   |   |   |— main.go         # /health + /example endpoints
|   |   |   |   |   |— internal/
|   |   |   |   |   |   |— middleware/
|   |   |   |   |   |   |   |— auth.go  # (legacy) auth middleware
|   |   |   |   |   |   |   |— router/
|   |   |   |   |   |   |       |— init.go # (legacy) router init
|   |   |   |
|   |   |   |— middle/                 # Core feature service (:8082)
|   |   |   |   |
|   |   |   |   |— .air.toml
|   |   |   |   |— app/
|   |   |   |   |   |— main.go         # Loads SL graph, registers
|   |
|   |— /middleplaces
|   |   |
|   |   |— data/                      # Static GTFS data for SL transit
|   |
|   |— graph
|   |   |
|   |   |— sl_agency.csv
|   |   |— sl_routes.csv
|   |   |— sl_stop_times.csv
|   |   |— sl_stops.csv
|   |   |— sl_trips.csv
|   |   |
|   |   |— internal/
|   |   |   |
|   |   |   |— controller/
|   |   |   |   |— middle.go          # HTTP handler for /middleplaces
|   |   |   |
|   |   |   |— graph/
```

```

├── sl_graph.go    # In-memory SL transit graph
├── sl_data.go     # Loads GTFS CSVs into graph
├── sl_models.go   # GTFS data models
├── calculate.go   # Dijkstra shortest-path
├── search.go      # Graph search helpers
├── slapi.go       # SL Journey Planner API client
├── edge.go        # Graph edge model
├── vertex.go      # Graph vertex model
├── errors.go      # Graph-specific errors
├── middle/
├──   └── middle.go    # Average() and Median() midpoint
calc
├── places/
├──   └── nearby.go    # Google Places API + Overpass/OSM
fallback
├──   └── priority_queue/
├──     └── model.go    # Min-heap for Dijkstra
├── test/
├──   ├── geocoding_test/
├──     └── geocoding_test.go
├──   ├── graph_test/
├──     └── graph_test.go
├──   ├── middle_test/
├──     └── middle_test.go
├── sl/                                     # SL trip search service (:8083)
├──   ├── .air.toml
├──   ├── app/
├──     ├── main.go          # /trip and /trips endpoints
├──   ├── internal/
├──     ├── controller/
├──       └── trip.go        # TripEndpoint handler
├──     ├── searchaddress/
├──       └── slsearchaddress.go # SL address search API client
├──   └── test/
├──     ├── controller_test/
├──       └── controller_test.go
├──     ├── search_test/
├──       └── slsearchaddress_test.go
├── signinverification/                # Google Sign-In token verifier
├──   ├── .air.toml
├──   ├── app/
├──     ├── main.go          # /verify-google-signin endpoint
├──   ├── internal/
├──     ├── controller/
├──       └── verifygooglesignin.go # HTTP handler
├──     ├── verifier/
├──       └── googleverifier.go    # Google ID token
validation

```

```

├── user/                                # User + friends service (:8084)
│   ├── app/
│   │   └── main.go                    # Registers all user/friend/home
├── endpoints
│   ├── internal/
│   │   ├── controller/
│   │   │   ├── login.go              # POST /login – Google Sign-In entry
│   │   │   ├── profile.go           # User profile endpoints
│   │   │   └── friends.go            # Friend request / accept / decline
│   │   ├── search.go                 # GET /search – find users
│   │   ├── googleauth/
│   │   └── googleauth.go             # Google OAuth helpers for user
│   ├── / list
│   ├── service
│   │   ├── repository/
│   │   │   ├── user_repo.go          # DB queries for users
│   │   │   └── friend_repo.go         # DB queries for friendships
│   │   ├── shared/                   # Code shared across all services
│   │   │   ├── db/
│   │   │   │   └── db.go              # PostgreSQL connection via sqlx
│   │   │   ├── models/
│   │   │   │   ├── user.go            # User struct
│   │   │   │   ├── location/
│   │   │   │   │   ├── location.go    # Point and Address structs
│   │   │   │   │   └── geocoding/
│   │   │   │   │       └── nominatim.go # Nominatim geocoding response model
│   │   │   └── router/
│   │   │       ├── standard_router.go  # Init() and Start() – mux setup,
│   │   │       ├── auth_middleware.go  # Bearer token validation,
│   │   │       ├── UserClaims context injection
│   │   │       └── rate_limiter.go      # Per-IP tokenbucket rate limiter

```